ENVIRONMENT & ECOLOGY

59. The term 'M-STrIPES' is sometimes seen in the news in the context of

(a) Captive breeding of Wild Fauna
(b) Maintenance of Tiger Reserves
(c) Indigenous Satellite Navigation System
(d) Security of National Highways

Ans : (b) Monitoring system for tiger's Intensive protection and Ecological status is a software monitoring system launched by the Indian Government in 2010 in some tiger reserves to reduce the vulnerability of Tigers. The system would enable field managers to assist intensity and Spatial Coverage of patrols in a geographic information system (GIS) domain.

60. In India, if a species of tortoise is declared protected under Schedule I of the Wildlife (Protection) Act, 1972, what does it imply?

(a) It enjoys the same level of protection as the tiger.
(b) It no longer exists in the wild, a few individuals are under captive protection and now it is impossible to prevent its extinction.
(c) It is endemic to a particular region of India.
(d) Both (b) and (c) stated above are correct in this context.

Ans : (a) A number of wild animal and bird species have been listed in various schedules of the Indian wildlife (protection) Act, 1972. The Act provides for the protection of wild animals, birds and plants and for matters connected there with or ancillary there to. It has six schedules which give varying degrees of protection. Animals under schedule I of the Wildlife Act enjoy the highest protection in India since the tiger is also covered in schedule I; therefore, option (a) is correct.

61. Recently there was a proposal to translocate some of the lions from their natural habitat in Gujarat to which one of the following sites?

(a) Corbett National Park
(b) Kuno Palpur Wildlife Sanctuary
(c) Mudumalai Wildlife Sanctuary
(d) Sariska National Park

Ans : (b) An environmental ministry expert committee has approved Kuno Palpur in Madhya Pradesh as the second home for Asiatic lions found only in Gir National Park.
In 2013 Supreme Court ruled in favours of translocation of the Asiatic lions to Kuno in the interest of the genetic stability of the species. However, the Gujarat Government will not share lions unless 33 studies as mandated by international wildlife watchdog IUCN is completed.

62. With reference to 'Global Climate Change Alliance', which of the following statements is/are correct?

1. It is an initiative of the European Union.
2. It provides technical and financial support to targeted developing countries to integrate climate change into their development policies and budgets.
3. It is coordinated by World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD).

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 3 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans : (a) The GCCA was established by the European Union (EU) in 2007 to strengthen dialogue and cooperation with developing countries, in particular, least developed countries (LDCs) and small island developing states (SIDS). Statement 1 is correct.
The GCCA also provides technical & financial support to targeted developments countries to integrated climate change into their development policies and budgets Statement 2 is correct.
Under the second pillar, the GCCA+ acts as a source of technical and financial support for the worlds' climate - vulnerable countries whose populations need climate finance the most. International organizations, notably the centre for International Forestry Research.
World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD) are not partners of GCCA; other institutions like FAO, UNDP etc. are part of it. Therefore Statement 3 is incorrect.

63. Biological Oxygen Demand (BOD) is a standard criterion for

(a) Measuring oxygen levels in blood
(b) Computing oxygen levels in forest ecosystems
(c) Pollution assay in aquatic ecosystems
(d) Assessing oxygen levels in high altitude regions

Ans : (c) Biochemical Oxygen Demand (BOD), also called Biological oxygen demand, is the amount of dissolved oxygen needed (i.e. demanded) by the aerobic biological organism to break down organic material present in a given water sample at a certain temperature over a specific time period. The BOD value is most commonly expressed in milligrams of oxygen consumed per litre of sample during 5 days of incubation at 20ºC and is often used as a gauge of the effectiveness of waste water treatment plants. It is hyped as a conventional pollutant in the US clean water Act.

64. With reference to the role of UN–Habitat in the United Nations programme working towards a better urban future, which of the statements is/are correct?

I. UN–Habitat has been mandated by the United Nations General Assembly to promote socially and environmentally sustainable towns and cities to provide adequate shelter for all.
2. Its partners are either governments or local urban authorities only.
3. UN–Habitat contributes to the overall objective of the United Nations system to reduce poverty and to promote access to safe drinking water and basic sanitation.

Select the correct answer using the code given below:

(a) 1, 2 and 3 (b) 1 and 3 only
(c) 2 and 3 only (d) 1 only

Ans : (b) UN-Habitat is the United Nations programme working towards a better urban future. It is mandated by the UN General Assembly in 1978 and aims to promote socially and environmentally sustainable human settlements development and the achievement of adequate shelter for all. Through drawing together cooperation with committed partners, relevant stakeholders, and urban actors including at all levels of government as well as the private sector, UN-Habitat is applying its technical expertise to the New Urban Agenda and sustainable Development. So, statement 2 is incorrect.

65. If you want to see gharials in their natural habitat, which one of the following is the best place to visit?

(a) Bhitarkanika Mangroves
(b) Chambal River
(c) Pulicat Lake
(d) Deepor Beel

Ans : (b) National Chambal Sanctuary, also called the National Chambal Gharial wildlife sanctuary, is a tri state- protected area in northern India for the critically endangered Gharial (small crocodiles), the red crowned roof turtle and the endangered Ganges river dolphin.

66. Which of the following practices can help in water conservation in agriculture?

1. Reduced or zero tillage of the land
2. Applying gypsum before irrigating the field
3. Allowing crop residue to remain in the field

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (c) 1. No till farming (also called zero tillage or direct drilling) is a way of growing crops or pasture from year to year without disturbing the soil through tillage.
2. Major benefits of gypsum are: Source of calcium and sulfur for plant nutrition, Improves Saline and alkaline soils and treats aluminum toxicity, improves soil structure. Hence statement 2 is incorrect.
3. Residue from the crops lie on the surface of the field cooling it and increasing the moisture.

67. From the ecological point of view, which one of the following assumes importance in being a good link between the Eastern Ghats and the Western Ghats?

(a) Sathyamangalam Tiger Reserve
(b) Nallamala Forest
(c) Nagarhole National Park
(d) Seshachalam Biosphere Reserve

Ans : (a) Sathyamangalam Tiger Reserve:- It is located in the Erode district of Tamil Nadu and is part of Western Ghats and Eastern Ghats. Hence the option 1 is correct.
Sathyamangalam forest range is an important wildlife corridor in Niligiri Biosphere Reserve.
It is a genetic link between the four other protected areas which it adjoins including the Billigiriranga Swamy Temple wildlife sanctuary Sigur plateau, Mudumalai National Park, and Bandipur National Park.
Nallamala Forest: It is located in Kodagu district and Mysore district of Karnataka and is also called Rajiv Gandhi National park.
Seshachalam Biosphere Reserve:
It is located in the Southern region of Andhra Pradesh in the Eastern Ghats. This reserve is a habitat of famous red sanders and slender lories species.

68. Consider the following statements in respect of Trade Related Analysis of Fauna and Flora in Commerce (TRAFFIC):

1. TRAFFIC is a bureau under United Nations Environment Programme (UNEP).
2. The mission of TRAFFIC is to ensure that trade in wild plants and animals is not a threat to the conservation of nature.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans : (b) TRAFFIC was established in 1976 by IUCN and WWF, it is a wildlife trade monitoring network, and the leading non-governmental organisation working globally on trade in wild animals and plants in the context of biodiversity conservation and sustainable development. TRAFFIC is governed by the TRAFFIC committee, a steering group composed of members of TRAFFIC'S partner organizations WWF and IUCN. It is not a bureau under UNEP.

69. In the context of solving pollution problems, what is/are the advantage/advantages of bioremediation technique?

1. It is a technique for cleaning up pollution by enhancing the same biodegradation process that occurs in nature.
2. Any contaminant with heavy metals such as cadmium and lead can be readily and completely treated by bioremediation using microorganisms.
3. Genetic engineering can be used to create microorganisms specifically designed for bioremediation.

Select the correct answer using the code given below :